

RESEARCH INTERESTS PLANNING, ROBOTICS, FLEET MANAGEMENT, PARALLEL COMPUTING

Profile

PhD researcher in AI and robotics working on LLM-assisted planning, task and motion planning, and executable temporal plans for robotic systems. Experience with ROS/ROS2, simulation, optimization, logic programming, and research software development. Interested in physical AI systems that connect model training, validation, and real-world robot deployment.

Education

- Nov 2022 – Current **PhD in Information Engineering and Computer Science**, *University of Trento*, Italy
Topics: Multi-Agent Planning, Temporal Planning, Industrial Robotics.
Goal: Develop a holistic system that through large language models and logic programming is able to plan and verify schedules for industrial robots and execute them, learning from experience.
GitHub: idra-lab/PLANTOR
- Oct 2018 – Jul 2022 **Master's Degree in Computer Science**, *University of Trento*, Italy, Final mark: 109
Thesis title: "Comparison of Multi-Agent Path Finding Algorithms for an Industrial Scenario."
Thesis argument: managing a fleet of AGVs in a human populated environment.
Topics: AGV control, robotics principles, path and goal planning, fleet control.
Other strong acquired knowledge:
 - Machine learning and deep learning (Tensorflow and PyTorch);
 - Real time operating systems;
- Sep 2015 – Oct 2018 **Bachelor's Degree in Computer Science**, *University of Trento*, Italy
"Implementation of GPU algorithms for robot path planning."
Topics: CUDA GPU programming, robot motion planning, comfort control.

Work Experience

- Mar 2025 – Nov 2025 **Research Internship**, *Cognitive Robotics Department – DLR*, Munich, Germany
Worked on integrating task and motion planning for the generation of executable temporal plans for multi-agent robotic systems. Focused on connecting symbolic task-level reasoning with motion primitives, optimization, and execution-oriented validation.
Topics: Task and Motion Planning (TAMP), motion primitives, temporal planning, LLMs, optimization, robotic execution.
- Sept 2019 – Dec 2019 **Computer Scientist**, *CreateNet – FBK*, Trento, Italy
Worked on cutting-edge technologies for control and optimization of agricultural irrigation in a deployed system.
Topics: C programming language, LoRaWAN infrastructure, electronic sensor and actuators.
- Jan 2019 – Jul 2019 **High School Teacher**, *ITT Buonarroti-Pozzo*, Trento, Italy
Taught computer science to high school students:
1st year: mainly problem solving skills; 2nd year: basics of C programming.

Research Experience

- Sep 2022 – Oct 2022 **Research Fellowship – "Predoc"**, *University of Trento*, Italy
Topics: Multi-Agent Path Finding, fleet management
Goal: Creation of a framework encompassing different MAPF algorithms for testing and scalability analysis
- Dec 2020 – May 2021 **Research Project as Student**, *University of Trento*, Italy
 - Research on **Dubins** curves for optimal control of vehicles;
 - Implementation on **GPU** of dynamic programming for the multi-point Dubins problem;
 - Energetic analysis** of different solutions from embedded systems to server based ones.

Conference Organization

- September 2026 **Local chair**, *Next Generation AI Systems (NGEN-AI 2026)*, Trento, Italy
- September 2026 **Local chair**, *The International Conference on Intelligent Digital Forensics and Cybersecurity (IDFC 2026)*, Trento, Italy

September 2026 **Local chair**, *The International Conference on Quantum Computing Infrastructure and Technologies (QUANCOM 2026)*, Trento, Italy

Summer/Winter Schools

26th – 31st May 2024 **ICAPS 12th Summer School**, *Banff Centre for Arts and Creativity*, Canada
Summer school on task and motion planning.

12th – 16th Feb 2024 **EUSCOTHA: EUregio School on Control Theory and Applications**, *University of Trento*, Italy
Winter school on control theory and applications.

Teaching Experience

Fall 2024 [TA] **Robot Planning & its Applications**, *University of Trento*, Italy
Role: Class lectures, laboratory exercises, and exams.

Fall 2023 [TA] **Robot Planning & its Applications**, *University of Trento*, Italy
Role: Class lectures, laboratory exercises, and exams.

Fall 2023 [Tutor] **Programming 101**, *University of Trento*, Italy
Role: Extra exercises and clarifications for Programming 101 course.

Fall 2022 [TA] **Real Time Operating Systems and Middlewares**, *University of Trento*, Italy
Role: Class lectures and exams.

Software and Research Artifacts

PLANTOR A framework for task and motion planning with LLMs and logic programming.
GitHub: [idra-lab/PLANTOR](https://github.com/idra-lab/PLANTOR)

MDP Generator A framework for generating MDPs through logic programming and large language models.
GitHub: [idra-lab/prolog_mdp](https://github.com/idra-lab/prolog_mdp)

MPDP A framework to solve multi-point motion planning problems exploiting Dynamic Programming for Dubins and Reeds-Shepp.
GitHub: [icosac/MPDP](https://github.com/icosac/MPDP)

MAOF A framework for testing and comparing different Multi Agent Path Finding (MAPF) algorithms.
GitLab: [chaff800/MAOF](https://gitlab.com/chaff800/MAOF)

Publications

- [10] M. Frego, E. Saccon, D. De Martini, and L. Palopoli, "Complexity Reduction of the Three-Point Dubins Problem (3PDP) Via Symmetry Exploitation for Machine Learning Purposes," *2026 International Conference on Robotics and Automation (ICRA)*, 2026.
- [9] E. Saccon, D. De Martini, M. Saveriano, E. Lamon, L. Palopoli, and M. Roveri, "Automated Generation of MDPs Using Logic Programming and LLMs for Robotic Applications," *IEEE Robotics and Automation Letters*, vol. 11, no. 2, pp. 1770–1777, 2026. DOI: 10.1109/LRA.2025.3643276.
- [8] P. Pastorelli, S. Dagnino, E. Saccon, M. Frego, and L. Palopoli, "Fast shortest path polyline smoothing with G^1 continuity and bounded curvature," *IEEE Robotics and Automation Letters*, vol. 10, no. 4, pp. 3182–3189, 2025. DOI: 10.1109/LRA.2025.3540531.
- [7] E. Saccon and M. Frego, "A Machine Learning Approach for the Three-Point Dubins Problem (3PDP)," *Symmetry*, vol. 17, no. 12, 2025, ISSN: 2073-8994. DOI: 10.3390/sym17122133. [Online]. Available: <https://www.mdpi.com/2073-8994/17/12/2133>.
- [6] E. Saccon, A. Tikna, D. D. Martini, E. Lamon, L. Palopoli, and M. Roveri, *A temporal planning framework for multi-agent systems via llm-aided knowledge base management*, 2025. arXiv: 2502.19135 [cs.AI]. [Online]. Available: <https://arxiv.org/abs/2502.19135>.
- [5] E. Saccon, A. Tikna, D. De Martini, E. Lamon, L. Palopoli, and M. Roveri, "When prolog meets generative models: A new approach for managing knowledge and planning in robotic applications," in *2024 IEEE International Conference on Robotics and Automation (ICRA)*, IEEE, 2024, pp. 17 065–17 071.

- [4] E. Saccon, "Multi-agent open framework: Developing a holistic system to solve mapf (student abstract)," in *Proceedings of the International Symposium on Combinatorial Search*, vol. 16, 2023, pp. 198–199.
- [3] E. Saccon, L. Palopoli, and M. Roveri, "Comparing multi-agent path finding algorithms in a real industrial scenario," in *AlxIA 2022–Advances in Artificial Intelligence: XXIst International Conference of the Italian Association for Artificial Intelligence, AlxIA 2022, Udine, Italy, November 28–December 2, 2022, Proceedings*. 2023, pp. 184–197. DOI: 10.1007/978-3-031-27181-6_13.
- [2] E. Saccon, P. Bevilacqua, D. Fontanelli, M. Frego, L. Palopoli, and R. Passerone, "Robot Motion Planning: can GPUs be a Game Changer?" *2021 IEEE 45th Annual Computers, Software, and Applications Conference (COMPSAC)*, pp. 21–30, 2021. DOI: 10.1109/COMPSAC51774.2021.00015.
- [1] M. Frego, P. Bevilacqua, E. Saccon, L. Palopoli, and D. Fontanelli, "An Iterative Dynamic Programming Approach to the Multipoint Markov-Dubins Problem," *IEEE Robotics and Automation Letters*, vol. 5, no. 2, pp. 2483–2490, 2020. DOI: 10.1109/LRA.2020.2972787.

Conference Presentations

- Jun 2026 **Poster, Vienna, Austria**
International Conference on Robotics and Automation 2026 (ICRA26)
Presented the poster for the IEEE R-AL paper "Automated Generation of MDPs Using Logic Programming and LLMs for Robotic Applications".
Presented the poster for the IEEE ICRA paper "Complexity Reduction of the Three-Point Dubins Problem (3PDP) Via Symmetry Exploitation for Machine Learning Purposes".
- Oct 2025 **Speaker, Hangzhou, China**
International Conference on Intelligent Robots and Systems 2025 (IROS25)
Presented the IEEE R-AL paper "Fast shortest path polyline smoothing with G^1 continuity and bounded curvature".
Presented the work titled "Planning with Natural Language for Task-Oriented Robots" at the SPACE workshop.
- Sep 2025 **Speaker, Rende, Italy**
International Conference on Logic Programming 2025 (ICLP25)
Presented the work "When Prolog Meets Generative Models: a New Approach for Managing Knowledge and Planning in Robotic Applications – Abstract" at the CALRA workshop.
- Sep 2024 **Speaker, Santiago De Compostela, Spain**
40th European Conference on Artificial Intelligence (ECAI 2024)
Presented the work "Adaptive and Scalable Knowledge Management for Robotic Applications via Probabilistic Logic Languages" at the Adjustable Autonomy and Physical Embodied Intelligence AAPEI 24 workshop during ECAI24.
- Jun 2024 **Speaker, Banff, Canada**
34th International Conference on Automated Planning and Scheduling 2024 (ICAPS24)
Presented the paper for the Previously Published Paper track: "When Prolog Meets Generative Models: A New Approach for Managing Knowledge and Planning in Robotic Applications"
Presented the extended abstract for the Doctoral Consortium: "Adaptive and Scalable Knowledge Management for Robotic Applications via Probabilistic Logic Languages"
- May 2024 **Speaker, Tokyo, Japan**
International Conference on Robotics and Automation 2024 (ICRA24)
Presented the main paper: "When Prolog Meets Generative Models: A New Approach for Managing Knowledge and Planning in Robotic Applications"
- Jul 2023 **Speaker, Prague, Czech Republic**
16th International Symposium on Combinatorial Search 2023 (SoCS23)
Presented the extended abstract for the Doctoral Consortium: "Multi-Agent Open Framework: Developing a Holistic System to Solve MAPF"
- Nov 2022 **Speaker, Udine, Italy**
21st International Conference of the Italian Association for Artificial Intelligence (AlxIA 2022)
Presented the conference paper: "Comparing Multi-Agent Path Finding Algorithms in a Real Industrial Scenario"
- Jul 2021 **Speaker, Madrid (virtual), Spain**
IEEE COMPSAC 2021 Intelligent and Resilient Computing for a Collaborative World
Presented the conference paper: "Robot Motion Planning: can GPUs be a Game Changer?"

Academic Service

Reviewing **IEEE R-AL, IEEE ICRA 2026, ICAPS-25, SoCS-25**
Program Committee **AAAI 2027, CARLA 2026 @ FLOC26, ICAPS 2026, PAIS 2025**

Awards and Grants

Jun 2024 **ICAPS24 DC Support**, Banff, Canada. As part of the DC, I was awarded a support of ~ 300\$ to register at the ICAPS24 conference.
Jul 2023 **SoCS23 Travel Grant**, Prague, Czech Republic. I was awarded a travel grant of 400 euros.

Skills

AI Planning Multi-agent planning, task and temporal planning, TAMP, MAPF, MDPs, STNs, logic programming, probabilistic reasoning.
Robotics ROS(1,2), Gazebo, MoveIt, motion planning, AGV fleet management.
Optimization OR-Tools, IBM CPLEX, constraint optimization, scheduling, dynamic programming.
LLMs and ML Azure OpenAI SDK, Hugging Face Transformers, vLLM, PyTorch, TensorFlow, prompt-based knowledge extraction, LLM evaluation pipelines.
Programming Python, C, C++, Prolog, MATLAB, Bash, JavaScript.
Software Tools Git, CUDA, Linux, Django, Node.js.

Communication and Interpersonal Skills

Good teamworking skills learned through various projects assigned during university courses and participations in Hackathon events (Hackathon Italia 2017, Hackathon FBK 2016, Hackathon Google 2018) and Google Hashcode (in 2019, 2020, 2021, 2022).

Languages

English Full professional knowledge

Professional References

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Date: July 17, 2026

Enrico Saccon